SpecNet: A Deep Learning Assistant for Analog & Mixed-Signal (AMS) Circuit Design

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Introduction

Analog and mixed-signal (AMS) circuit design is a manual and iterative process that relies heavily on designer intuition and SPICE simulation, creating a bottleneck in early-stage design [1]. This paper introduces SpecNet: a deep learning model that maps performance specifications to transistor dimensions. The goal of the project is to narrow the space around the optimal sizes, thus reducing the amount of simulation required and accelerating the design flow. This project involves development of two models: an artificial neural network (SpecNet-ANN) to leverage physical/device relationships, and a graph neural network (SpecNet-GNN) to exploit the locality between circuit nodes.

For the purpose of this project, the scope is limited to only one circuit family (single-stage operational amplifiers). Inputs to this model are specific design goals: DC gain, phase margin, and gain-bandwidth product, and outputs are symmetrically-matched transistor widths. Both models will be compared against a linear model that simulates design tools used in industry. Figure 1 presents the project roadmap in more detail.

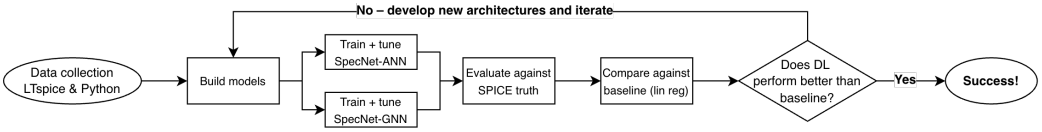


Figure 1: Project Roadmap

Background and Related Work

To demonstrate ongoing research in this topic, three academic papers are presented. Also presented are two startups to show that strides in AI for circuit design have grown beyond academia. Unlike the RL-based approaches below that require iterative simulator feedback, SpecNet aims to learn a direct mapping from specifications to device dimensions, enabling faster inference.

1. *"Automated Deep Learning Platform for Accelerated Analog Circuit Design"* (ISOCC 2022). Proposes a deep learning-based optimization loop where neural networks learn circuit performance from simulation data and enable fast analog sizing [4].
2. *"An Open-Source AMS Circuit Optimization Framework Based on Reinforcement Learning"* (IEEE Access, 2024). Introduces an RL framework that optimizes AMS circuits from specifications to layouts by learning from simulator feedback [5].
3. *"Automated Design of Analog Circuits Using Reinforcement Learning"* (IEEE TCAD, 2022). Demonstrates the viability of machine learning to tune analog circuit parameters, showing that RL can replace manual iterative design [8]
4. *Cognichip*, building 'Artificial Chip Intelligence' to assist in digital design. Their focus is on applying artificial intelligence to create novel circuits [3].
5. *ChipAgents*, building AI agents to accelerate the chip verification process. Their focus is on design automation and acceleration [2].

Data Processing

Data collection began with designing an operational amplifier in LTspice, using a predetermined common-mode voltage and resistance value to establish an operating point. A SPICE script was used to increment through the sample space and record values; all measurements were saved to a log file from which they were extracted using a Python script. The dataset was z-standardized, then split into training, validation, and testing subsets. Table 1 shows a complete example measurement being parsed to decimal form and saved as a CSV row.

Dataset Element	Extracted Form	Conversion	Saved Form
DC Gain	(122.435 dB, 0°)	N/A	122.435 dB
PM	(38.8787 dB, 0°)	$10^{38.8787/20}$	$87.89°$
GBW	(157.115 dB, 0°)	$10^{157.115/20}$	71738121 Hz
W1	N/A	N/A	0.00025 m
W3	N/A	N/A	0.00025 m

Table 1: LTspice Log File to CSV Values

A short regression test on the outputs provided an R^2 value of 0.98 and 0.99 for W1 and W3 respectively, but only 0.69 for W5, suggesting there are other factors that affect W5. Indeed, the size of the current mirror transistors is heavily dependent on the bias current (a fixed value) and not the inputs [7]. As a result, M5/M6 sizing predictions were removed from the model.

Between the limitations of LTspice and the circuit family itself, challenges were imposed on the data collection process, which had implications for the overall project. The three most significant challenges are detailed below:

1. *LTspice only steps 3 parameters* – The proposal called for 6 inputs to the model, consisting of 3 widths and length values each [6]. However, because LTspice only allows for 3 parameters to be iterated over, a common length was set with only transistor widths modified.
2. *SPICE parameters impose size limits* – The SPICE library used is meant for discrete circuit simulation, which have significantly larger dimensions than integrated circuits. Although the goal was to simulate sizes in the low-μm to nm range, the SPICE library limited **accurate** simulation to the mm to high-μm range.
3. *Phase margin changes very little* – Phase margin does not change as drastically as expected; the range is 9.27 degrees with all values above 80 degrees, an excellent margin of safety.

Model Architecture

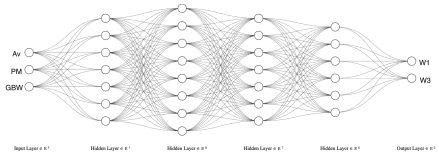
SpecNet-ANN is a five-layer artifical neural network (with layer sizes 3-128-256-128-64-2). The important hyperparameters of the model are: trained using Adam (lr = 0.001), MSE loss (as this is a regression problem), batch size of 32, and LeakyReLU activations (model learns in all ranges).

SpecNet-GNN is a graph convolutional neural network built on three GCN layers followed by a two-layer fully-connected regression head. The circuit is represented as a graph where each node corresponds to a transistor with its width as the scalar node feature, and edges encode the connections between transistors. The model takes this graph as input and uses global max pooling to collapse it into a single representation. The important hyperparameters of the model are: trained using Adam (lr = 0.001), MSE loss, batch size of 32, and LeakyReLU activations.

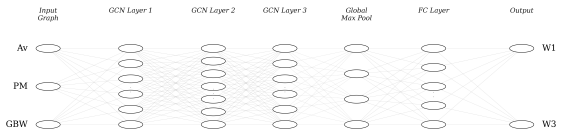
Figure 2 below shows the network architecture for both implemented models.

Baseline Model

The baseline model for this project is a multivariate linear regression model, implemented in PyTorch as a single linear layer. This model is meant to replicate industry practice, where designers use linear models to find approximate ranges; they then run SPICE simulations on points in these ranges to find the optimal sizes.



(a) SpecNet-ANN



(b) SpecNet-GNN

Figure 2: Architecture Diagrams for Both Implemented Models

Results

Accuracy for these models is measuring using a tolerance-based approach: a predicted set of parameters is considered accurate if the relative error – the absolute difference between the predicted and true output, normalized by the true output – falls within some threshold. This metric directly reflects the models' intended application: in circuit design, a sizing estimate does not need to be exact, but it needs to be close enough to the optimum to significantly reduce the simulation required. Equation 1 below provides an expression for this method of evaluation.

$$\text{Model accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\frac{|\hat{y}_i - y_i|}{y_i} \leq \varepsilon\right] \tag{1}$$

where y_i is the true size, $\hat{y}_i$ is the predicted size, ε is the tolerance threshold, and N is the total number of samples evaluated. $\mathbf{1}$ represents the indicator function, which evaluates to 1 if the condition inside the parentheses is true and 0 otherwise.

Accuracy was measured on a testing dataset not previously used to train the models, and additional samples. These additional samples were generated using a new SPICE sweep over a different parameter range, ensuring they were not used in any stage of model selection or training. Figure 3 compares both SpecNet models against the baseline for variance tolerance values. Each point on the graph represents what portion of predicted transistor dimensions was within that tolerance. Specific numerical values are provided in Table 2.

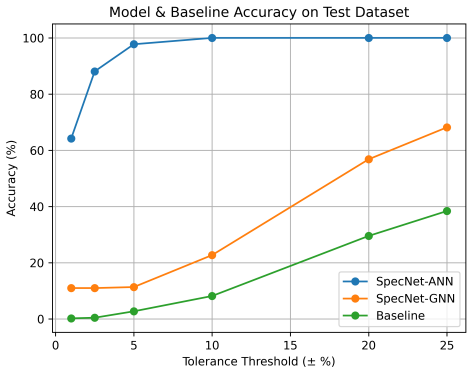


Figure 3: Tolerance-Accuracy Curves

Tolerance	ANN	GNN	Baseline
1	64.20	11.00	0.23
2.5	88.07	11.11	0.45
5	97.73	11.36	2.73
10	99.99	22.73	8.18
20	99.99	56.82	29.55
25	99.99	68.18	38.41

Table 2: Numerical Results (all in %)

Discussion

First, observe that SpecNet-GNN performs modestly better than the baseline, especially at higher tolerances. While far from perfect, it nonetheless helps designers narrow the simulation space. SpecNet-ANN performs the best at every tolerance and is clearly the superior model; over the testing dataset, over 95% of its predictions were within 5% of the labels. In comparison, the GNN and baseline achieved only 22% and 11% respectively.

The motivation behind developing two models in parallel was to evaluate whether the relationship between transistor sizes and select performance characteristics is more closely tied to device/physics

equations or circuit topology. The results in Figure 3 support the conclusion that operational amplifier behaviour is dominated more strongly by the nonlinear device/physics equations, which the ANN can successfully learn as a direct input-output mapping.

The GNN's comparatively poorer performance is likely attributed the circuit topology being fixed. When every sample shares the exact same op-amp structure, the graph's edge connectivity carries no inherent encoding across samples. The GNN is then rediced to learning from node features alone, forfeiting its structural advantage.

Ethical Considerations

The primary ethical consideration for this project surrounds intellectual property and access. During training, it is expected that the model inherently encodes highly-protected process parameters in its weights and biases. If the model is leaked, it may expose manufacturers as adversaries could prompt it with known values to reverse-engineer these parameters. Thus, it is recommended to run this model locally over a secure network (rather than through cloud providers) and restrict access to engineers working in high-level front-end design.

A secondary consideration is the risk of over-reliance. A model that performs well in training could produce confident but incorrect size estimates for specifications that fall outside of its training set; an engineer unfamiliar with the model might accept incorrect estimates without paying attention to the underlying limitations. This is especially consequential in analog design, where a poor initial sizing can propagate through many simulation iterations before it is detected. It is recommended that predictions be sanity-checked against known constraints and treated as an informed starting point rather than a definite, unconditional solution.

Conclusion

This work introduces SpecNet, a deep learning framework for accelerating AMS design by mapping performance specifications directly to transistor dimensions. The motivation behind this project was to reduce the reliance on iterative SPICE simulations and manual tuning, which poses a significant bottleneck in early-stage design.

The results demonstrate that this approach is effective within the scoped problem. In particular, the SpecNet-ANN model achieves strong performance across all tolerance thresholds, with over 95% of predictions falling within 5% of ground truth sizes. This suggests that, for fixed-topology circuits such as single-stage operational amplifiers, the relationship between specifications and dimensions can be learned as a nonlinear mapping. In contrast, the weaker performance of SpecNet-GNN highlights that graph-based approaches provide limited benefit when circuit topology remains constant.

Future work should focus on expanding both the dataset and the modeling approach. An important improvement is to extend the framework to multiple circuit topologies, where GNNs are expected to better leverage topological variety between designs. Incorporating additional design parameters, such as transistor lengths and bias conditions, would also improve the model's completeness. Finally, integrating SpecNet into a larger closed-loop system for desginer ease-of-use could bridge the gap between estimation and industry-acceptable design workflows.

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